YZM2022: Data Mining

Homework 3

Recommendation System Project

Amazon Reviews 2023 — Musical Instruments

Tuğana Öykü Yıldız
24018020

Project Link: (https://github.com/oykutugana/recommendation-system-musical-instruments)

Summary: This project builds and compares recommendation systems for the Amazon Musical Instruments category. Content-based filtering, collaborative filtering, and hybrid models are implemented. All models are evaluated with 5-fold cross validation.

1. Objective

This project builds a recommendation system for the Amazon Reviews 2023 dataset. Only the **Musical Instruments** category is used. The dataset is very large, so it is sampled first. The main goals of the project are listed below:

- Perform Exploratory Data Analysis (EDA).
- Build a Content-Based recommendation system.
- Build Collaborative Filtering models.
- Develop Hybrid Recommendation architectures.
- Compare all approaches with the same metrics.
- Analyze the effect of review text, ratings, metadata, and product images.
- Sentence-BERT is used for semantic text embeddings (in addition to TF-IDF).
- CLIP is used for image embeddings.
- A sentiment-weighted rating formula is tested: 0.7 × rating + 0.3 × sentiment.
- 9 content-based combinations are tested (3 models × 3 strategies).
- 8 CF experiments are run (4 models × 2 rating types).

2. Dataset

The Amazon Reviews 2023 dataset is used. Only the **Musical Instruments** category is selected. The full data is too large, so sampling is applied first.The full Musical Instruments category contains ~3M interactions. The data is sampled to 473,647 interactions using a date filter (2014-2023) and iterative 5-core filtering.

Sampling steps

- **Time filter: 2014-01-01 to 2023-09-06 (~9.7 years).**
- **5-core filtering:** Each user and each item must have at least 5 reviews. This step is repeated until the data is stable.
- **Cold-start hold-out: 10,000 users with 1-2 reviews are held out before 5-core filtering is applied to the main set. After 5-core filtering, 6,904 of these users have interactions with items remaining in the final item space.**

The sampled dataset is uploaded to Hugging Face. The code loads it from there: https://huggingface.co/datasets/oyku-tugana/amazon-musical-instruments-2014-2023-5core

Dataset statistics

Metric	Value
Interactions	473,647
Unique users	53,127
Unique items	23,158
Sparsity	99.96%
Mean rating	4.45
Date range	2014-01-01 to 2023-09-06
Items with image	99.98% (23,153 / 23,158)
Items with price	69.2%
Cold-start users	6,904 (7,753 interactions)

3. Exploratory Data Analysis

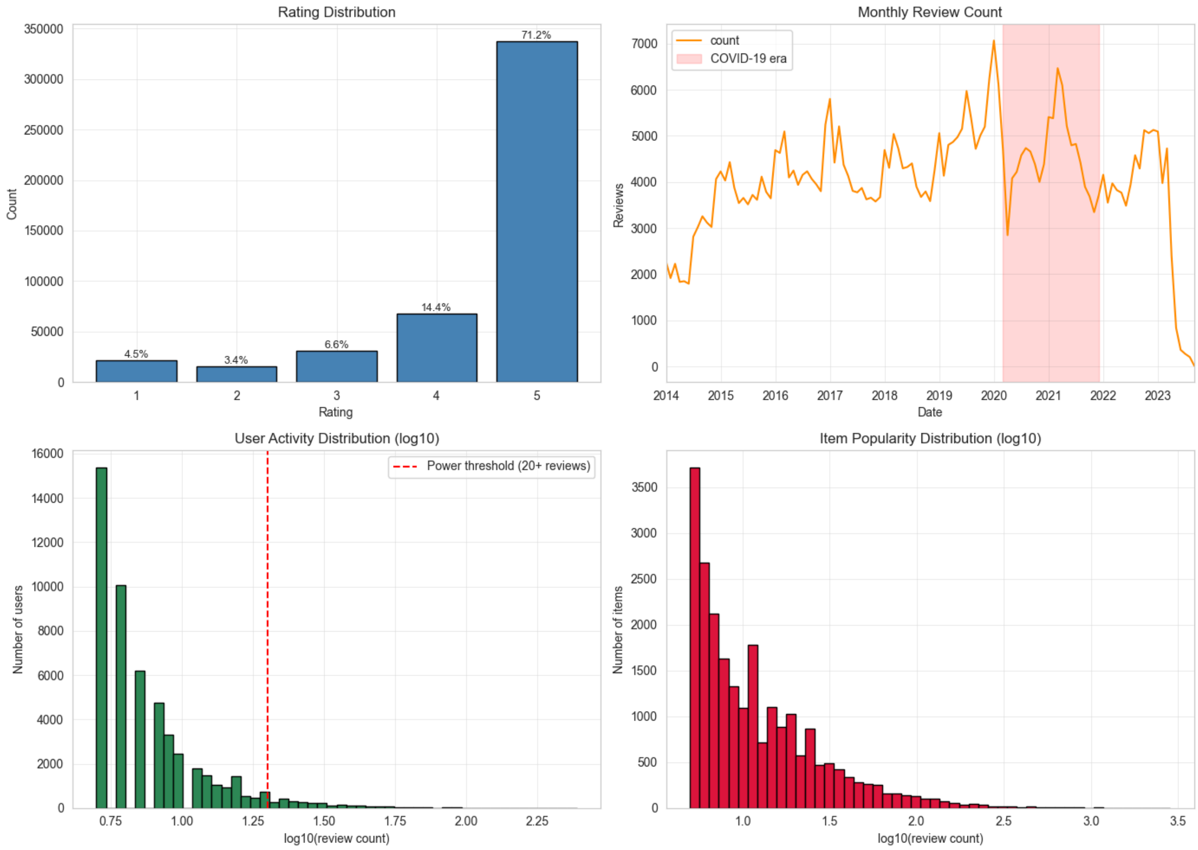

Figure 1. eda_overview.png

- *Top-left: rating distribution is heavily skewed toward 5 stars (71.2%).*
- *Top-right: monthly review count drops sharply after mid-2023 due to incomplete recent data collection; COVID-19 era (2020-2021) is highlighted.*
- *Bottom-left: user activity is long-tailed; only a small fraction (power users, 20+ reviews) accounts for most reviews.*
- *Bottom-right: item popularity is also long-tailed.*

3.1 User Analysis

User activity is long-tailed. The top 5% of users ("power users") give many reviews. These power users give slightly higher mean ratings (4.47 vs 4.43) but with higher variance. (std 0.81 vs 0.75), meaning they hold stronger and more varied opinions

3.2 Item Analysis

Item popularity is also long-tailed. Many items have few reviews. A few items are very popular. The most common product categories are stands, straps, picks, and cables. Prices vary widely.

3.3 Temporal Analysis

Monthly review counts peak around 2020 (yearly ~56K reviews) and then drop sharply toward the end of the collection window (2023).The reason is the data collection method — recent months are not fully collected. For this reason, the **last 12 months** are used as the test set, not the last 3 months.

3.4 Text Analysis

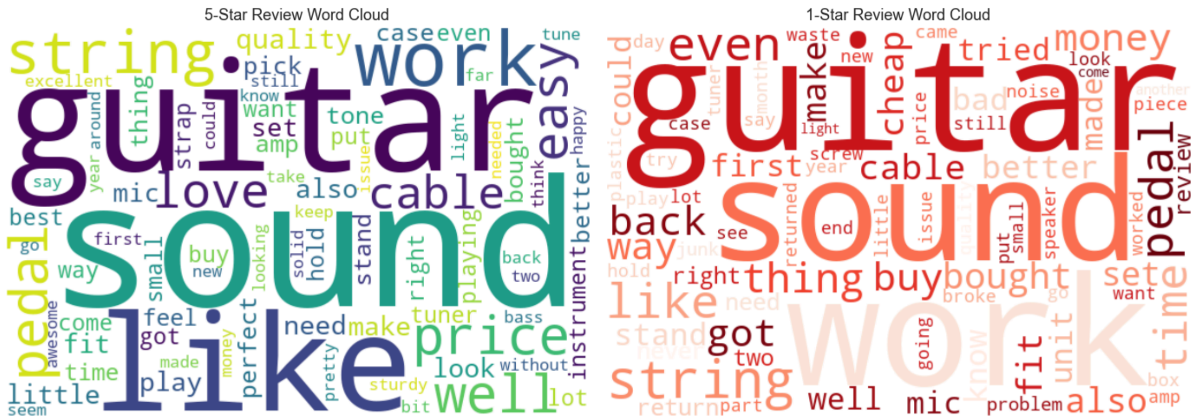

Figure 2. wordclouds.png

Word clouds for 5-star (left, green) and 1-star (right, red) reviews.
Common words appear in both clouds (guitar, sound, string, work) because review topics are similar across ratings.
The sentiment difference comes from context, not vocabulary — this is one reason VADER struggles with ratings in this category.

Sentiment analysis:

- VADER is used to compute sentiment for every review
- Important finding: ~19.78% of 1-star reviews have a positive VADER sentiment
- Reason: VADER reads words like "love" or "perfect" but misses context
- Example: "I love the design BUT it broke in one week" → VADER misses the "BUT"
- **The proposed solution:** rating and sentiment are mixed together. `adjusted_rating = 0.7 × rating + 0.3 × scaled_sentiment`. This idea is tested later with CF models (Section 4.2).

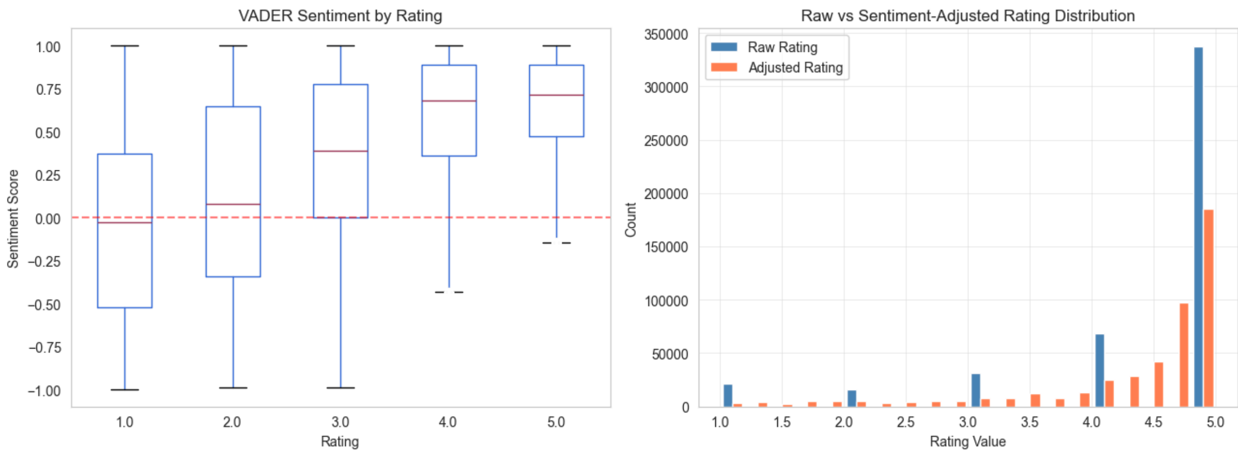

Figure 3. sentiment_analysis.png

Left: VADER sentiment score by star rating — sentiment is positively correlated with rating but with very wide variance, especially in low-rating reviews.

Right: raw rating vs sentiment-adjusted rating distribution. The adjusted rating smooths the 5-star peak and shifts mass to mid-range values.

4. Recommendation Models

4.1 Content-Based Filtering

Content-based filtering recommends items that look like items the user liked before. Three different content representations are tested:

- **TF-IDF: Sparse text vectors from product titles, descriptions, and reviews. Size: 23,158 items × 5,000 features. Cosine similarity is used.**
- **Sentence-BERT** (all-MiniLM-L6-v2): Dense semantic embeddings, 384 dimensions per item.
- **CLIP** (ViT-B/32): Dense image embeddings from product photos, 512 dimensions per item.

For each model, 3 user profile strategies are tested: *mean_liked* (mean of liked items), *last_n* (mean of last 5 items), and *itemknn_last* (items similar to the last one). This gives **9 combinations** in total.

Result: TF-IDF with last_n wins (NDCG@10 = 0.00667). This is surprising because semantic models (SBERT, CLIP) usually beat TF-IDF. Here, product titles like "Fender Stratocaster Electric Guitar Sunburst" are already very informative at the word level. CLIP is the weakest: visual similarity does not match functional similarity in this category.

4.2 Collaborative Filtering

Collaborative filtering uses past user-item interactions, not content. 4 models are built, each with 2 rating types (raw rating and sentiment-adjusted rating). This gives **8 experiments** in total.

Model	Type	NDCG@10 (raw)	NDCG@10 (sent)
User-CF	Memory-based	0.00805	0.00810 (+0.67%)
Item-CF	Memory-based	0.00116	0.00148 (+28.22%)
SVD	Latent factor	0.00760	0.00749 (−1.48%)
ALS	Implicit feedback	0.01069	0.01058 (−1.02%)

Key findings:

- **ALS is the best CF model (0.01069 in raw, 0.01058 in sentiment).**
- **Item-CF is the weakest.** The standard formula with sum-of-similarities in the denominator is used. This formula removes popularity bias, but in this category popularity bias was actually a useful signal.
- **Sentiment-adjusted rating shows a model-specific effect. Item-CF improves +28.22% with sentiment, but Item-CF is already the weakest model (raw NDCG@10 = 0.00116) so the absolute improvement is small. User-CF, SVD, and ALS show changes between -1.48% and +0.67%, all within noise. The proposed signature idea did not generalize across most CF families.**

The user mean rating distribution shows that the mass is concentrated between 4.0 and 5.0 (Figure 4). This narrow distribution is one reason rating-centering techniques (such as Pearson similarity) cannot extract additional signal in this dataset — most users behave similarly, so subtracting per-user means removes useful information instead of correcting for it. Cosine similarity was preferred for User-CF in this study.

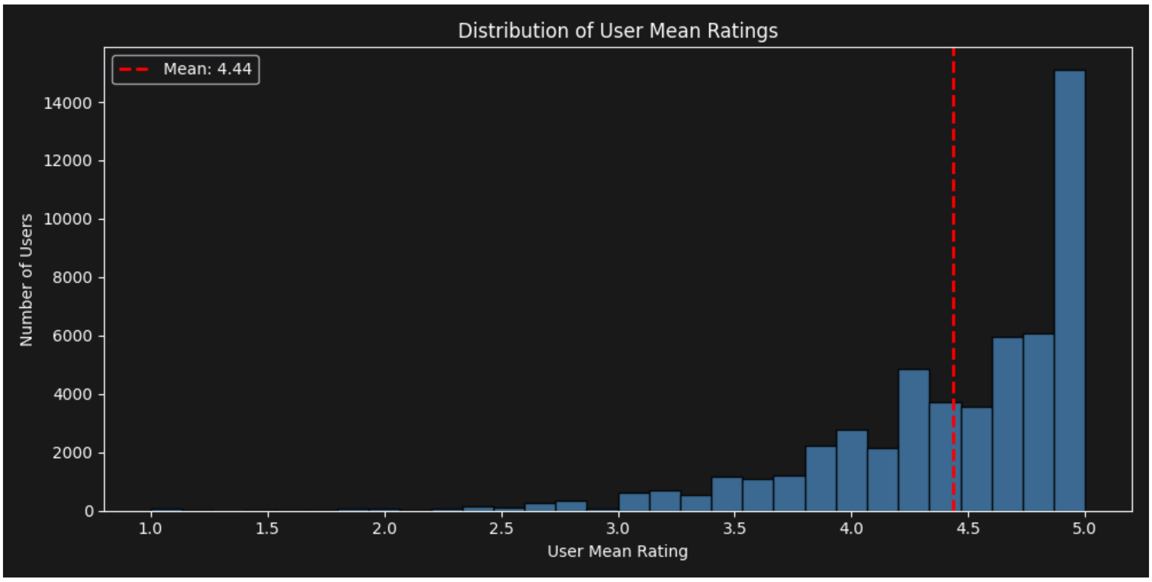

Figure 4. user_mean_rating_dist.png

The mass is concentrated between 4.0 and 5.0 (mean = 4.44, red dashed line). Few users have mean ratings below 3.0. This narrow distribution makes Pearson-style mean-centering ineffective in this dataset.

4.3 Hybrid Recommendation

Score = α × Score_CF + (1−α) × Score_CB. The value of α is swept from 0 to 1 in steps of 0.05.

α	NDCG@10	Coverage
0.00 (pure CB / TF-IDF)	0.00667	0.816
0.30	0.00858	0.796
0.50	0.01072	0.705
0.75 (best)	**0.01324**	0.343
0.90	0.01189	0.213
1.00 (pure CF / ALS)	0.01069	0.191

α = 0.75 is the best. It gives NDCG@10 = 0.01324. This is +23.9% better than pure ALS and +98.5% better than pure TF-IDF. The optimum has shifted further toward CF compared to smaller datasets because more interaction data strengthens the ALS signal.

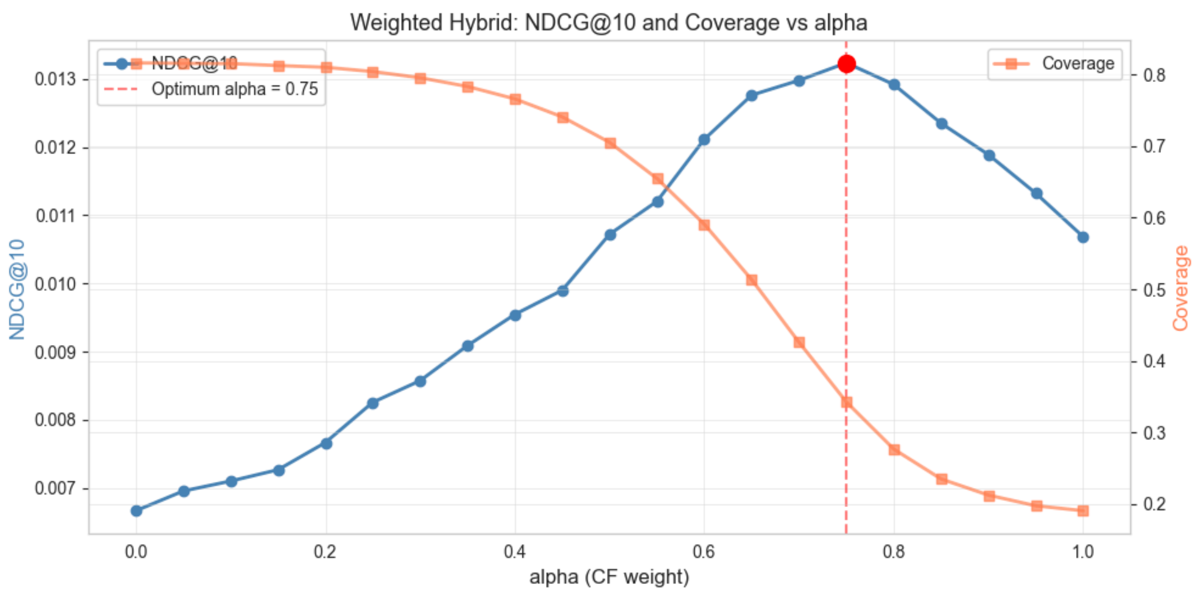

Figure 5. alpha_curve.png

NDCG@10 (blue) and Coverage (orange) as α changes from 0 (pure content-based) to 1 (pure ALS). NDCG@10 reaches its maximum at α = 0.75 (red marker, NDCG@10 = 0.01324). Coverage drops as α increases, because ALS has lower coverage than TF-IDF. The optimal α gives a good balance between accuracy and diversity.

Multimodal 4-way ensemble

A richer mix is also tested: TF-IDF + SBERT + CLIP + ALS, with a grid search for weights. The best weights from the fine grid (0.05 steps) are shown below:

Component	Weight
TF-IDF	0.25
Sentence-BERT	0.00
CLIP	0.00
ALS	0.75
Total NDCG@10	**0.01324 (identical to α=0.75 hybrid)**

CLIP weight is zero in all searches. Both the coarse grid (0.25 steps) and the fine grid (0.05 steps) found CLIP=0 in the temporal split. 5-fold cross validation later confirmed this: all 5 folds independently found the same {tfidf:0.25, sbert:0.0, clip:0.0, als:0.75} solution. The multimodal result (0.01324) is exactly equal to the simple α=0.75 hybrid (0.01324) because the multimodal optimum is mathematically identical to alpha-weighted hybrid with α=0.75. The simpler model is the recommended choice.

4.4 Feature Selection

Three feature selection techniques with different theoretical backgrounds are selected:

- **Chi-square (χ^2):** Frequency-based, fast, works on sparse matrices.
- **Mutual Information:** Information-theoretic measure.
- **ReliefF:** Instance-based, uses nearest neighbors.

Target (personalized): Binary classification — items liked by the user (rating ≥ 3.5) are positive examples; random unseen items are negative examples. This target is built only from training data (no leakage). Total samples: 728,364 with 50/50 class balance.Tested K values: 500, 1000, 2000 features. The reduced TF-IDF matrix is then used for content-based recommendation.

Technique	Best K	NDCG@10	Drop vs Full
Chi-square	2000	**0.00599**	**−10.11%**
ReliefF	2000	0.00587	−11.99%
Mutual Info	2000	0.00571	−14.31%
Full (baseline)	5000	**0.00667**	—

Result: No reduced version beats the full baseline, but Chi-square top-2000 comes close with only −10.11% drop. The new ranking is Chi² > ReliefF > MI (different from the earlier dataset where ReliefF was the best reduced version). Chi-square and Mutual Information selected brand names (addario, gls audio, snark, ernie ball, donner) and specific product features (analog delay, high carbon, steel core). ReliefF selected more general category words (guitar, bass, accessories, sound, tone). Chi² and MI overlap is high (399/500 features), but the overlap with ReliefF is lower (123 and 144 out of 500, three-way intersection 109). With the larger dataset (473K interactions, 23K items), brand-specific features turned out to be more useful for the personalized recommendation task than category-level features.

5. Evaluation

5.1 5-Fold Cross Validation

All major models are tested with 5-fold cross validation. Each fold uses 80% training and 20% test. Total training data: 473,647 interactions (each fold trains on ~378,917 interactions).. Hybrid weights (α and multimodal weights) are **re-optimized inside each fold** (no leakage from the temporal split).

Model	NDCG@10 (mean ± std)	Coverage
Hybrid α-weighted	**0.03435 ± 0.00128**	0.342
Hybrid multimodal	0.03428 ± 0.00130	0.337
ALS_raw	0.03097 ± 0.00097	0.183
SVD_raw	0.01809 ± 0.00067	0.022
TF-IDF	0.01234 ± 0.00090	0.810
Sentence-BERT	0.00830 ± 0.00078	0.661

Important CV finding — CLIP contribution does not generalize:

- Temporal split coarse and fine grid both found CLIP weight = 0
- 5-fold CV optimized multimodal weights inside each fold (coarse grid for speed)
- **In all 5 folds, the optimal multimodal weight was** `{tfidf: 0.25, sbert: 0.0, clip: 0.0, als: 0.75}` — CLIP and SBERT got zero every time
- All searches across multiple data splits (temporal coarse, temporal fine, 5-fold CV) independently arrived at the same result, confirming that CLIP does not contribute in this category — a robust, generalizable finding
- **Lesson:** marginal findings must be verified with cross validation

Other CV findings:

- All models are stable — standard deviations are less than 3% of the mean
- Hybrid α-weighted and multimodal results are almost identical (difference 0.00007) — because the CV multimodal collapsed to a 2-component model (TF-IDF + ALS), mathematically identical to α-weighted with α=0.75
- CV values are ~2.6× higher than temporal split values → random k-fold is an easier task than predicting the future from the past

5.2 Rating Prediction Metrics (MAE, RMSE)

Only SVD and ALS produce rating-scale predictions, so these metrics are measured only on these two models.

Model	MAE (mean ± std)	RMSE (mean ± std)
SVD_raw	4.3120 ± 0.0038	4.4639 ± 0.0114
ALS_raw	4.3616 ± 0.0026	4.4486 ± 0.0019
Trivial baseline (predict global mean = 4.45)	0.8092	1.0927

- MAE values are very high (~5.3-5.4× higher than the trivial baseline)
- Reason: these models are not calibrated to predict rating values — they are designed for top-K ranking
- ALS works with a confidence scale, not a rating scale — computing MAE/RMSE on ALS is conceptually questionable.
- SVD predicts close to 0 for unseen items because the matrix is 99.96% sparse
- For the main use case (top-K recommendation), NDCG@10 and Precision@K are the right metrics

5.3 Cold-Start Evaluation

Two scenarios are tested with the cold-start users. Of the 10,000 held aside earlier, 6,904 have interactions in the final item space (1-review: ~6,055; 2-review: ~849):

Scenario	Method	NDCG@10	vs Populari ty
1-review users	Popularity baseline (only option)	0.00891	—
2-review users	Popularity	0.01735	—
2-review users	CB hybrid (TF-IDF + SBERT + CLIP)	**0.02067**	**+19.09%**

Findings:

- 2 reviews are enough to do better than popularity
- Content-based hybrid beats popularity by +19% (the smaller margin compared to the earlier dataset is because popularity itself is now stronger with more interaction data)
- CLIP is again the weakest single model (NDCG@10 = 0.01066, but still below TF-IDF and SBERT)
- Consistent with the warm-phase finding that visual signal does not contribute in this category

5.4 Trivial Baseline Lift

To confirm that the models really learn (and are not just lucky), the models are compared with two trivial baselines:

Model	NDCG@10	Lift vs Random
Random	0.00023	1.0×
Popularity	0.00857	37.7×
TF-IDF	0.00667	29.3×
ALS_raw	0.01069	46.9×
Hybrid (best)	**0.01324**	**58.2×**

All models are **far above random**. The best hybrid is also clearly above popularity, so the system shows real learning, not just popularity bias.

6. Discussion

This section answers the 6 questions..

6.1 Best Performing Recommendation Architecture

- **Best model: α-weighted hybrid with α = 0.75 (75% ALS + 25% TF-IDF) at temporal split**
- **In 5-fold CV: α-weighted hybrid (per-fold optimal α) gives NDCG@10 = 0.03435 ± 0.00128 — about +10.9% better than ALS alone**
- A more complex 4-way multimodal ensemble (TF-IDF + SBERT + CLIP + ALS) gives a similar score (0.03428 ± 0.00130)
- The difference is too small to be statistically meaningful
- Inside CV, the multimodal optimization collapsed SBERT and CLIP weights to zero → mathematically identical to α-weighted hybrid
- **Recommendation:** the simpler α-weighted hybrid — fewer parameters, less complexity

6.2 Advantages and Disadvantages of Each Method

Method	Pros	Cons
Content-Based Filtering (TF-IDF, SBERT, CLIP)	• Works for new items (item cold-start) • Explainable • High coverage (82% for TF-IDF) • Less popularity bias	• Filter bubble (user only sees similar content) • Poor for new users (user cold-start) • Depends on text/image quality • CLIP fails here: visual ≠ functional similarity
Collaborative Filtering (User-CF, Item-CF, SVD, ALS)	• Uses "crowd wisdom" • Does not need text or images • Can find surprising recommendations	• Fails on cold-start (no history = no recommendation) • Popularity bias • Low coverage (ALS: 19%, SVD: 2%) • Memory-based methods are slow at scale
Hybrid Methods	• Combines strengths of both paradigms • Better accuracy and diversity • Partial cold-start solution	• More complex (many models to manage) • More parameters to tune (α-sweep: 21 runs; multimodal grid: 80 runs)

6.3 Computational Complexity

Component	Complexity	Real time (laptop CPU)
TF-IDF vectorization	O(n × text_len)	~5 s
SBERT embeddings (GPU)	O(n × dim)	~3 min (Colab T4)
CLIP embeddings (GPU)	O(n × image)	~5 min (Colab T4)
ALS training (15 iter)	$O(k^2 \times (u+i) + nnz \times k)$	~30 s
α-sweep (21 values)	O(α × u × i)	~31 s
Multimodal grid (110 runs)	~110 × hybrid eval	~2.2 min
5-fold CV (full pipeline)	5 × one fold	~25 min
ReliefF feature selection	$O(n^2 \times f)$	~4 min (with subsample)

The expensive parts (image embeddings, ReliefF, full CV) are computed **once**. After that, prediction is fast (less than 100 ms per user).

6.4 Scalability Challenges

The current scale of this project: ~53k users, ~23k items, 474k interactions. The whole pipeline fits on a laptop.

If the system is scaled up to 1M users and 100K items:

- **ALS training:** scales linearly
- **Prediction per user:** ~50 ms, still acceptable.
- **Score matrix in memory:** 1M × 100K × 4 bytes = **400 GB**. Not possible to store. Batch processing becomes necessary.

Solutions for production:

- **Approximate Nearest Neighbor (Faiss, Annoy, HNSW)** — reduces k-NN search to O(log n).
- **Reduce latent factors** — k=32 instead of k=64 → ~4× faster.
- **Incremental ALS** — partial update for new interactions, no full retraining.
- **Caching** — store top-K for popular users.
- **Distributed computing** — Spark MLlib, PyTorch Distributed.

6.5 Cold-Start Problem

Cold-start is the situation when a new user (or new item) has no interaction history. Pure CF fails completely in this case.

The cold-start tests in this project show:

- **1-review users: Only popularity works. NDCG@10 = 0.00891.**
- **2-review users: Content-based hybrid reaches NDCG@10 = 0.02067 — +19% better than popularity.**

Suggested strategy for production deployment:

- **0-1 interactions:** popularity + a few random items for exploration.
- **2-5 interactions:** content-based blend, gradually add CF.
- **5+ interactions:** full warm pipeline (hybrid ALS + TF-IDF).

6.6 Effect of Textual and Visual Features

Textual features (TF-IDF, Sentence-BERT) — strong in this category:

- TF-IDF alone: NDCG@10 = 0.00667
- TF-IDF gets 25% of the weight in the best multimodal model (along with 75% ALS)
- SBERT receives 0% weight in the new dataset (the larger search space confirms it adds no value here)
- Reason: product titles already carry rich information at word level (e.g., *"Fender Stratocaster Electric Guitar Sunburst"* gives brand, category, color, shape)
- Word matching is enough

Visual features (CLIP) — weak in this category:

- CLIP solo: NDCG@10 = 0.00233 (only ~35% of TF-IDF)
- All searches (temporal coarse, temporal fine, 5-fold CV) found CLIP = 0
- **5-fold CV confirmed this — in all 5 folds, optimal CLIP weight was 0 (and SBERT was also 0)**
- This is a robust finding across multiple data splits — CLIP does not contribute in this category

Hypothesis:

- A black electric guitar and a black classic guitar look similar to CLIP (color, shape, composition)
- But for the user, these are very different products (different music styles)
- **Visual similarity ≠ functional similarity** in this domain

Important note:

- This finding is **category-specific**
- In other categories (clothing, furniture, decoration), visual features may be much more important
- CLIP should not be added everywhere without testing first
- Marginal findings from a single data split should always be confirmed with cross validation

6.7 Limitations

This study has the following methodological limitations:

1. MAE/RMSE are not the right metrics for ALS. ALS is designed for top-K ranking, not rating prediction. Its MAE values (~4.37) are reported ,but conceptually these metrics are not meaningful for ALS. SVD is a better candidate for rating prediction, but even it is ~5.3-5.4× worse than the trivial baseline because the model is optimized for top-K.

2. Hyperparameter optimization is limited. KNN k=50, SVD components=50, ALS factors=64 are set as standard values from the literature. No systematic grid search is done on these. Better results may be possible.

3. Feature selection is done outside the cross-validation loop. The target is built only from train data (no temporal leakage), but feature selection is not repeated inside each CV fold. A small optimistic bias may remain.

4. Random K-Fold is used instead of TimeSeriesSplit. Time-series data ideally needs Out-of-Time validation. In this project, the temporal split is the primary evaluation; CV is used only as a stability check. The fact that CV values are ~2.6× higher than temporal values is direct evidence of this leakage. CV results should not be interpreted as true performance estimates.

5. CLIP weakness has two possible explanations. Either visual similarity ≠ functional similarity in musical instruments (category-specific), or zero-shot pretrained CLIP is generally weak for recommendation. Testing on visual-first categories (clothing, furniture) is needed to separate the two hypotheses.

6. Single-dataset evaluation. All findings are specific to Musical Instruments. Generalization to other Amazon categories (Books, Electronics, Beauty) or other platforms (Movielens, Yelp) is not tested.

7. Conclusion

Multiple recommendation system architectures are built and tested on the Amazon Musical Instruments dataset:

- **15+ model variants**
- **3 feature selection techniques**
- **5-fold cross validation** for statistical reliability

Main lessons:

1. Hybrid models are the best

- α-weighted hybrid (0.75 × ALS + 0.25 × TF-IDF) gives +10.9% over the best single CF model in CV

2. Simple beats complex

- The 4-way multimodal ensemble is not statistically better than the simple 2-way α-hybrid
- Inside CV, multimodal even collapsed to a 2-component model
- Simple version is recommended

3. CLIP does not help here

- All temporal grid searches (coarse + fine): CLIP weight = 0
- 5-fold CV: CLIP weight = 0 in all 5 folds (independent verification)
- The CLIP finding is robust and generalizable across multiple data splits

4. Sentiment-adjusted rating did not generalize

- Only Item-CF showed clear sentiment benefit (+28.22%), but Item-CF was already the weakest model
- The other 3 CF models (User-CF, SVD, ALS) showed changes within noise (-1.48% to +0.67%)
- The proposed "signature" idea did not give clear wins

5. Feature selection trades accuracy for speed

- Features can be reduced by 60% with only 10-14% accuracy loss
- Chi-square (with personalized target) gives the best reduced version

6. 2 reviews are enough for cold-start

- Content-based hybrid beats popularity by 19% with just 2 user reviews (smaller margin than smaller datasets because popularity is stronger with more data)

Main lesson from this project: Theoretical advantages do not always work in practice, and marginal findings from a single data split must be verified with cross validation. CLIP embeddings (weight = 0 across all grid searches: temporal coarse, temporal fine, and 5-fold CV) and sentiment-adjusted rating (no consistent win across CF models) are concrete examples. Method choice must be tested empirically for each domain. Adding a sophisticated model without proper validation can hurt the system instead of helping it.

Project Structure

All work is contained in the main notebook. The folder structure is shown below:

```
TuganaOyku_Yildiz_24018020/
├── TuganaOyku_Yildiz_24018020.ipynb    (main notebook — runs everything)
├── README.md
├── requirements.txt
├── results/
│   ├── embeddings/                    (SBERT + CLIP, pre-computed on Colab GPU)
│   ├── figures/                       (PNG plots, generated by notebook)
│   └── tables/                        (JSON + Excel results, generated by notebook)
└── report_24018020.pdf                (this file)
```

How to run: the main notebook is opened in Jupyter, then Kernel → Restart → Run All is selected. Total runtime is about 70 minutes on a laptop CPU (CV takes ~25 minutes). The dataset is downloaded automatically from Hugging Face (oyku-tugana/amazon-musical-instruments-2014-2023-5core).

Note on embeddings: The SBERT and CLIP embeddings in *results/embeddings/* are pre-computed because they need GPU (Google Colab T4) and take about 8 minutes total to generate. The main notebook loads them directly, so it can run on a laptop CPU without GPU.